

Bentley® Software licensed to Connected User: P Kodanda RamaRao	Job No. EXNO 07	Sheet No. 1	Rev
Job Title PORTAL FRAME WITHOUT SWAY PFWS	Part	Ref	
Client	By V.UMA SRI	Date 8/9/2026	Chd
File PORTAL FRAME WITHOUT SWAY PFWS.std	Date Time 08-Sep-2026 15:27		

Job Information

	Engineer	Checked	Approved
Name:	V.UMA SRI		
Date:	8/9/2026		

Comments:	
Structure Type:	SPACE FRAME

Geometry

Entity Type	Count	Highest
Nodes	4	4
Analytical Members	3	3

Load Cases

Load Case Type	Count
Primary	2
Combination	1

Included in this printout are data for:

All	The Whole Structure
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Load Case Table

Included in this printout are results for load cases:

L/C	Type	Name
1	Primary	LIVE LOAD
2	Primary	DEAD LOAD
3	Combination	COMBINATION LOAD CASE 3

Node Displacements

Node	L/C	X (in)	Y (in)	Z (in)	Result ant (in)	rX (rad)	rY (rad)	rZ (rad)
1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	3	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	1	0.002	-0.001	0.215	0.215	0.003	0.000	0.000
	2	0.000	-0.001	0.000	0.001	0.000	0.000	0.000
	3	0.025	-0.021	2.686	2.686	0.034	0.000	-0.002

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Node Displacements Cont...

Node	L/C	X (in)	Y (in)	Z (in)	Result ant (in)	rX (rad)	rY (rad)	rZ (rad)
3	1	0.002	-0.001	0.215	0.215	0.003	0.000	0.000
	2	0.000	-0.001	0.000	0.001	0.000	0.000	0.000
	3	0.022	-0.020	2.686	2.686	0.034	0.000	0.001
4	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	3	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Nodes

Node	X (m)	Y (m)	Z (m)
1	0.000	0.000	0.000
2	0.000	4.000	0.000
3	6.000	4.000	0.000
4	6.000	0.000	0.000

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	1	2	4.000	1	0.000
2	2	3	6.000	1	0.000
3	3	4	4.000	1	0.000

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
3	COMBINATION LOAD CASE 3	1	LIVE LOAD	12.500
		2	DEAD LOAD	12.500

Node Displacements

Node	L/C	X (in)	Y (in)	Z (in)	Result ant (in)	rX (rad)	rY (rad)	rZ (rad)
1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2	0.000	0.000	0.000	0.000	0.000	0.000	0.000

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Node Displacements Cont...

Node	L/C	X (in)	Y (in)	Z (in)	Result ant (in)	rX (rad)	rY (rad)	rZ (rad)
1	3	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	1	0.002	-0.001	0.215	0.215	0.003	0.000	0.000
	2	0.000	-0.001	0.000	0.001	0.000	0.000	0.000
	3	0.025	-0.021	2.686	2.686	0.034	0.000	-0.002
3	1	0.002	-0.001	0.215	0.215	0.003	0.000	0.000
	2	0.000	-0.001	0.000	0.001	0.000	0.000	0.000
	3	0.022	-0.020	2.686	2.686	0.034	0.000	0.001
4	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	3	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Beam Displacement Summary

Type	Beam	L/C	d (m)	X (in)	Y (in)	Z (in)	Result ant (in)
Max X	3	3	1.000	0.043	-0.015	1.511	1.511
Min X	3	2	0.000	0.000	-0.001	0.000	0.001
Max Y	3	3	4.000	0.000	0.000	0.000	0.000
Min Y	3	3	0.000	0.022	-0.020	2.686	2.686
Max Z	1	3	4.000	0.025	-0.021	2.686	2.686
Min Z	1	1	0.000	0.000	0.000	0.000	0.000
Max Rst	2	3	3.000	0.024	-0.172	2.686	2.691

Beam End Force Envelope

Sign convention is as per the action of the joint on the beam.

Beam	Node		Max +ve	L/C	Max -ve	L/C
1	1	FX (kN)	853.183	3	0.000	
		FY (kN)	0.000		-58.343	3
		FZ (kN)	0.000		0.000	3
		MX (kip-in)	0.000	3	0.000	
		MY (kip-in)	2,212.687	3	0.000	
		MZ (kip-in)	0.000		-583.804	3
	2	FX (kN)	0.000		391.129	3
		FY (kN)	-58.343	3	0.000	

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Beam End Force Envelope Cont...

Beam	Node		Max +ve	L/C	Max -ve	L/C
1	2	FZ (kN)	0.000	3	0.000	
		MX (kip-in)	0.000		0.000	3
		MY (kip-in)	0.000		2,212.687	3
		MZ (kip-in)	0.000		1,481.725	3
2	2	FX (kN)	58.343	3	0.000	
		FY (kN)	391.129	3	0.000	
		FZ (kN)	0.000	3	0.000	
		MX (kip-in)	0.000		-2,212.687	3
		MY (kip-in)	0.000		0.000	3
		MZ (kip-in)	1,481.725	3	0.000	
	3	FX (kN)	0.000		58.343	3
		FY (kN)	-176.952	3	0.000	
		FZ (kN)	0.000		0.000	3
		MX (kip-in)	0.000		2,212.687	3
		MY (kip-in)	0.000	3	0.000	
		MZ (kip-in)	0.000		1,326.579	3
3	3	FX (kN)	176.952	3	0.000	
		FY (kN)	58.343	3	0.000	
		FZ (kN)	0.000		0.000	3
		MX (kip-in)	0.000	3	0.000	
		MY (kip-in)	2,212.687	3	0.000	
		MZ (kip-in)	1,326.579	3	0.000	
	4	FX (kN)	0.000		639.006	3
		FY (kN)	0.000		58.343	3
		FZ (kN)	0.000	3	0.000	
		MX (kip-in)	0.000		0.000	3
		MY (kip-in)	0.000		2,212.687	3
		MZ (kip-in)	-738.950	3	0.000	

Beam Max Moments

Beam	Node A	Length (m)	L/C		d (m)	Max My (kip-in)	d (m)	Max Mz (kip-in)
1	1	4.000	1	Max + ve	0.000	177.015	4.000	35.287
				Max -ve	0.000	0.000	0.000	-7.492

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Beam Max Moments Cont...

Beam	Node A	Length (m)	L/C		d (m)	Max My (kip-in)	d (m)	Max Mz (kip-in)
1	1	4.000	2	Max + ve	0.000	0.000	4.000	83.251
				Max -ve	0.000	0.000	0.000	-39.213
			3	Max + ve	0.000	2,212.68 7	4.000	1,481.72 5
				Max -ve	0.000	0.000	0.000	-583.804
2	2	6.000	1	Max + ve	5.500	0.000	0.000	35.287
				Max -ve	0.000	0.000	1.000	-40.538
			2	Max + ve	0.000	0.000	0.000	83.251
				Max -ve	0.000	0.000	3.000	-85.664
			3	Max + ve	5.500	0.000	0.000	1,481.72 5
				Max -ve	0.000	0.000	2.500	- 1,281.07 8
3	3	4.000	1	Max + ve	0.000	177.015	0.000	22.875
				Max -ve	0.000	0.000	4.000	-19.903
			2	Max + ve	0.000	0.000	0.000	83.251
				Max -ve	0.000	0.000	4.000	-39.213
			3	Max + ve	0.000	2,212.68 7	0.000	1,326.57 9
				Max -ve	0.000	0.000	4.000	-738.950

Beam Combined Stresses Summary

Beam	L/C	Length (m)	Max Comp			Max Tens		
			Stress (psi)	d (m)	Corner	Stress (psi)	d (m)	Corner
1	1	4.000	369.394	4.000	3	- 339.472	4.000	1
	2	4.000	86.044	4.000	3	-65.540	4.000	1
	3	4.000	5,692.9 67	4.000	3	- 5,062.6 49	4.000	1
2	1	6.000	37.879	1.000	1	-35.932	1.000	3
	2	6.000	80.776	3.000	1	-75.201	3.000	3
	3	6.000	1,395.9 70	0.000	3	- 1,301.9 48	0.000	1
3	1	4.000	357.698	4.000	2	- 341.979	0.000	1

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Beam Combined Stresses Summary Cont...

Beam	L/C	Length (m)	Max Comp			Max Tens		
			Stress (psi)	d (m)	Corner	Stress (psi)	d (m)	Corner
3	2	4.000	86.044	0.000	3	-65.540	0.000	1
	3	4.000	5,379.1 46	0.000	2	- 5,093.9 82	0.000	1

Academic

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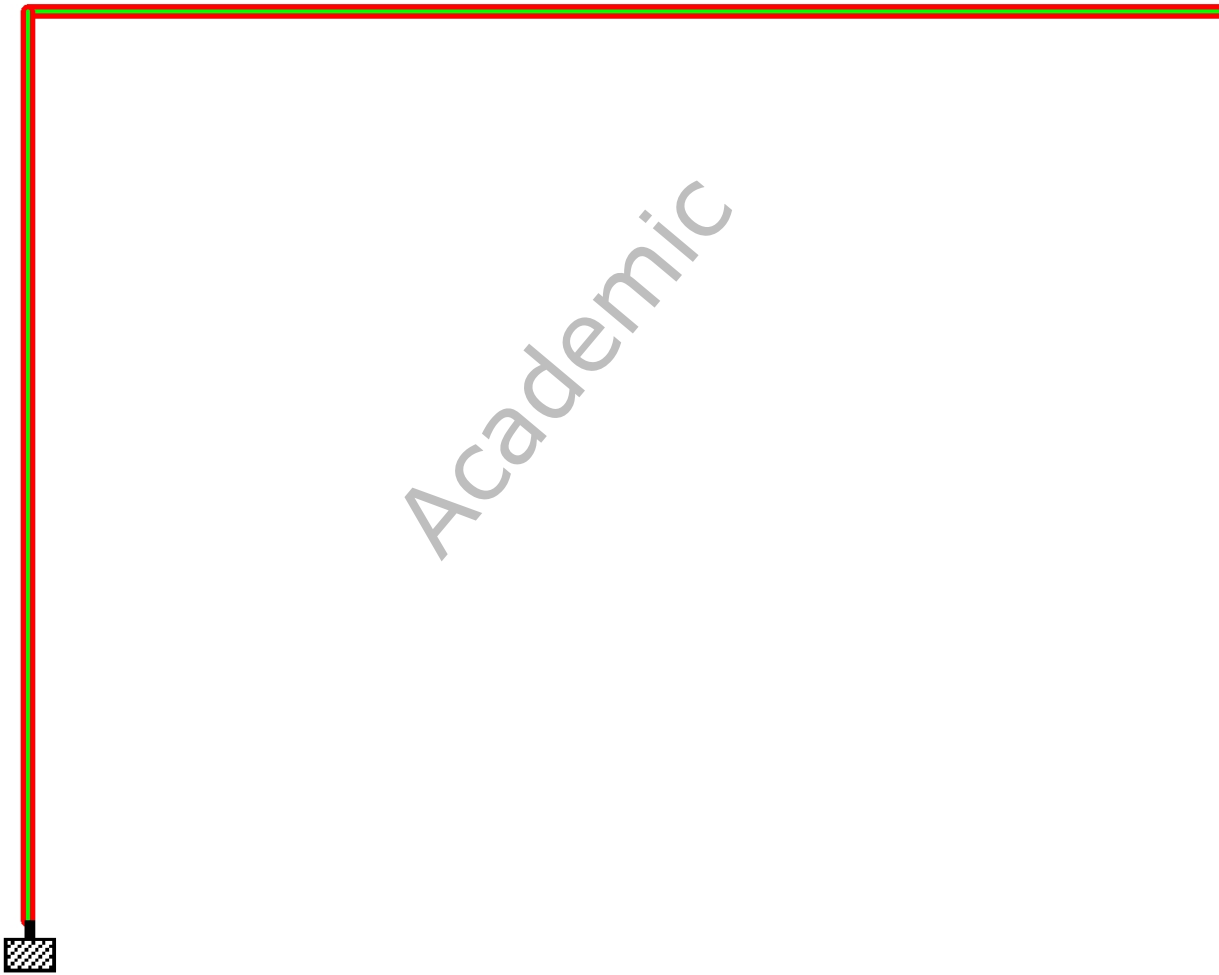
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y **V.UMA SRI**

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Load 2 : D

Whole Structure Displacements 39.3701in:1m 2 DEAD LOAD

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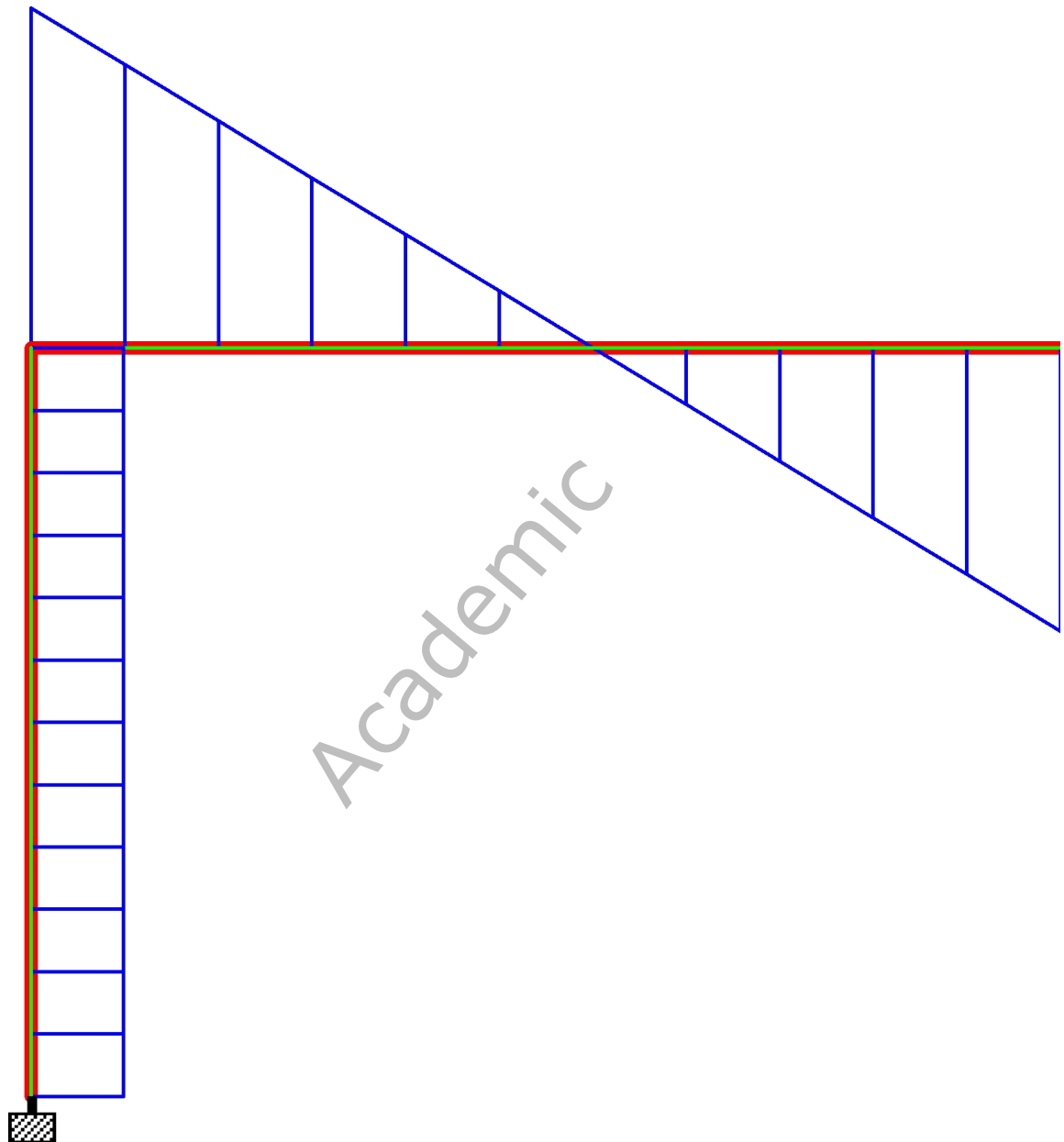
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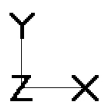
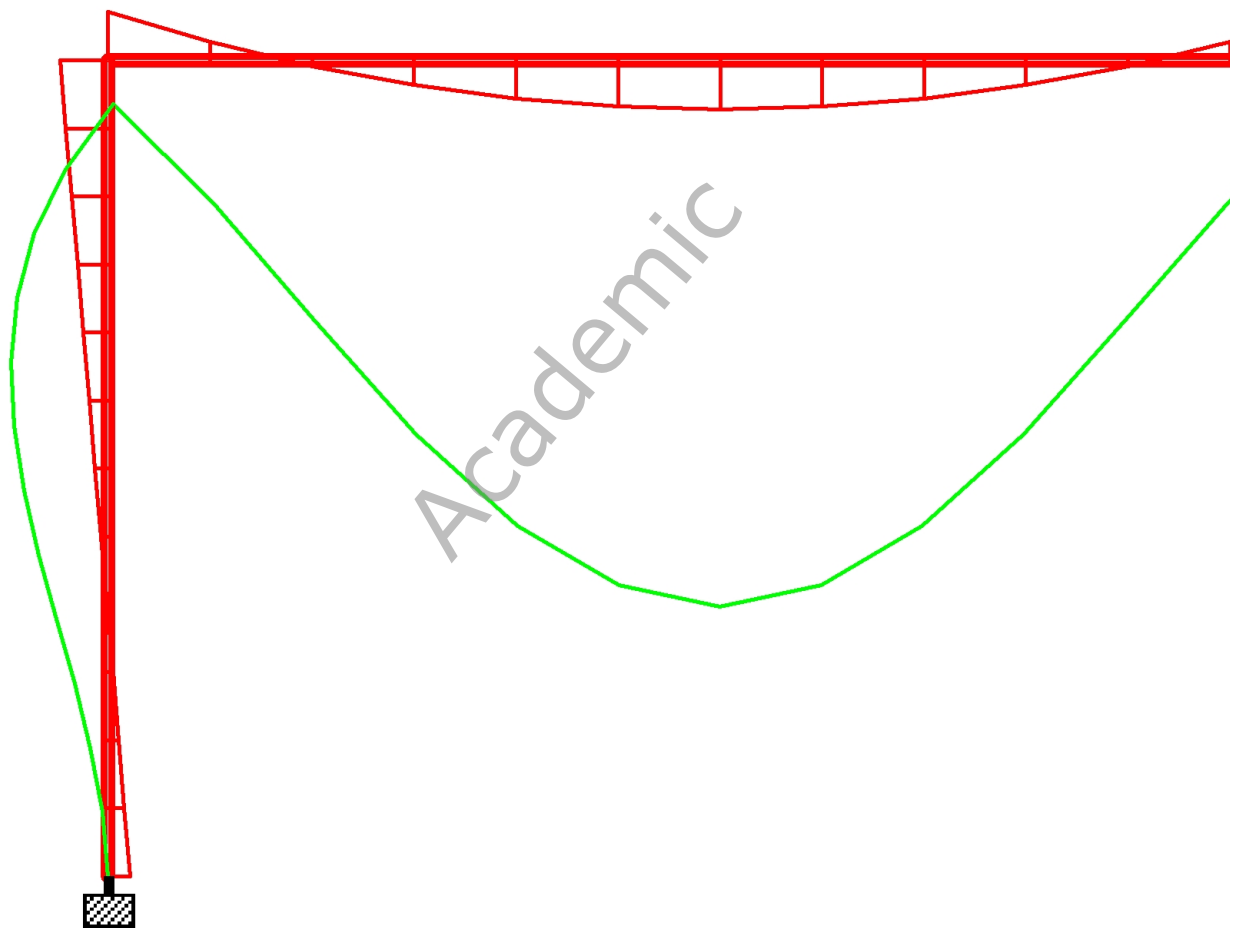
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Y
Z X

Load 2 : Shear Y : D

Whole Structure Fy 7.00507kN:1m Displacements 15.748in:1m 2 DEAD LOAD



Load 2 : Bending Z : D

Whole Structure Mz 354.331kip-in:1m Displacements 0.00393701in:1m 2 DEAD LOAD